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# Universidade de Aveiro

Departamento de Electrónica, Telecomunicações e Informática

42081 - Real-Time Operating Systems

Midterm lab exam - SAMPLE, $1^{st}$ Semester 26-27, Sept $30^{th}$ 2026

Duration: 2h45m

ID #: _____ Name: _____

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## AI model - No AI

This lab midterm exam assesses the individual's/group's ability to solve and justify problems independently.

The use of IA, the Internet, or external communication is not permitted, nor is the possession of any type of electronic device or unauthorized materials, even if turned off. Only the resources indicated in the instructions are permitted.

Possession or use of unauthorized items will result in the cancellation of the exam, in accordance with the REUA.

Student signature: _____

Student signature: _____

## Indications

- Students can use the provided API documentation in printed (paper) format. The sheet's empty space (both sides) can be freely used for annotations.
- Just one sheet of paper per group, properly identified before entering the room
- Students can (and should) use the man() pages for obtaining help on the use of syscalls
- Only the lab PC can be used. No other electronic devices are allowed.
- Besides the code, some questions may require you to copy results, copy commands or analyze results. You should create a Libreoffice Writer document and put the required information there. Identify clearly your answers. E.g.:
  Q2.D) The command used is: # stress-ng –matrix 1 -t 1m
- **Important:** This is a practical evaluation that intends to assess the capacity of students to, given a problem, create solutions, code them, find and fix bugs and deliver **working software**. Code that does not approach a working solution will not be graded.

1. Create a C program that establishes a deterministic periodic execution environment encapsulated in a thread. (5) points)

   A) (2) Create a program that creates a periodic thread activated every 100 ms. At the moment the thread just keeps an activation counter and prints it to the terminal whenever activated (1, 2, 3, ...)

   B) (2)Modify the program so that the thread above is assigned to a real-time scheduler class, with priority 50

   C) (1) Apply eventual additional features that can contribute positively to the execution determinism of the thread

2. Extend the previous C program to simulate a distributed data processing via Message Queues (15 points)

   A) (7) Create two additional threads, say Processor and Output, that shall be sporadic, triggered by data availability. The periodic task sends data (the counter, uint32_t type) to the Processor task. The controller task computes the moving average (window of 3) of the received values and sends it to the Output task, as a float. The Output task just outputs to the terminal the received value. Assign appropriate real-time classes and priorities to ensure the correct data flow: Periodic (source of data) → Processor → Output (sink of data).

   B) (1.5) Restrict all three application threads to execute on a single CPU core

   C) (0.5) Modify the thread's code to output the time of each activation, in the form "ss.mmm" (s: seconds, m:milliseconds). Let the program run for around 1 second (10 or more activations) and copy the timing information printed to the terminal to the answers' document.

   D) (1.5) Introduce background system load using the stress-ng. Copy the (complete) command you have used to the answers' document.

   E) (1.5) Let again the program run for around 1 second (10 or more activations) and copy the timing information printed to the terminal to the answers' document. Compare this output with the previous one and take conclusions (are the results the expected ones? Why? - Be objective. 3 to 4 lines of text at most.

   F) (2) Using the printed results sketch a Gantt diagram with the tasks execution, pinpointing activations and communication events

   G) (1) Use KernelShark to confirm the results of your Gantt chart sketch. Take a screenshot and save it with a suitable name.